Chunk influence on the RAG predictions

Labels: **judge**, sample tier: **all**. Quadrants: correct = |error| ≤ 0.5, wrong = |error| ≥ 1.0, texts in between = buffer (not sampled).

Party positions (CHES 2024 lrgen, 1-7 scale; = ground truth definition): DIE LINKE 1.85, BÜNDNIS 90/DIE GRÜNEN 2.83, SPD 3.08, BSW 3.17, FDP 4.63, CDU 4.95, CSU 5.54, AfD 6.56

1. Base rates (full population)

The sample has 25 texts per quadrant, so it over-represents rare quadrants (above all Q2). Pooled sample numbers below are weighted by these base rates; buffer texts are missing from the sample entirely.

model	texts	Q1 share	Q2 share	Q3 share	Q4 share	buffer share
llama_3b	710	0.09	0.19	0.38	0.11	0.23
llama_70b	775	0.03	0.09	0.15	0.23	0.49
llama_8b	774	0.09	0.03	0.26	0.20	0.41
mistral_14b	770	0.05	0.06	0.18	0.35	0.37
mistral_3b	761	0.10	0.15	0.29	0.13	0.33
mistral_8b	760	0.12	0.06	0.35	0.21	0.26
qwen_32b	775	0.10	0.05	0.29	0.12	0.44
qwen_3b	773	0.01	0.03	0.54	0.02	0.40
qwen_72b	775	0.07	0.04	0.23	0.17	0.49
qwen_7b	773	0.12	0.07	0.37	0.17	0.28
pooled	7646	0.08	0.08	0.30	0.17	0.37

Party composition per quadrant (share of the quadrant's texts)

Parties close to the scale centre are 'correct' for a constant centrist prediction, so quadrant membership partly reflects the party mix:

input_party	Q1	Q2	Q3	Q4	buffer	all texts	baseline correct (|err| ≤ 0.5)	position
DIE LINKE	0.12	0.12	0.31	0.10	0.28	0.23	0.15	1.85
BÜNDNIS 90/DIE GRÜNEN	0.14	0.27	0.18	0.21	0.21	0.20	0.35	2.83
SPD	0.29	0.28	0.20	0.27	0.35	0.28	0.33	3.08
FDP	0.14	0.08	0.09	0.04	0.06	0.07	0.22	4.63
CDU	0.07	0.03	0.05	0.06	0.07	0.06	0.31	4.95
CSU	0.01	0.01	0.02	0.00	0.00	0.01	0.10	5.54
AfD	0.23	0.21	0.16	0.31	0.04	0.15	0.48	6.56

constant prediction	counted correct for
1.00	—
1.50	DIE LINKE
2.00	DIE LINKE
2.50	BÜNDNIS 90/DIE GRÜNEN
3.00	BÜNDNIS 90/DIE GRÜNEN, SPD, BSW
3.50	SPD, BSW
4.00	—
4.50	FDP, CDU
5.00	FDP, CDU
5.50	CSU
6.00	CSU
6.50	AfD
7.00	AfD

2. Evidence use (full population, not selected on outcome)

R ~ B + E (+ model / + input party), E = mean position of the retrieved chunks' parties; SE clustered by text. The permutation null shuffles E across texts within each model (1000×). `naive_pull_corr` is the earlier corr(R − B, E − B); its null mean shows how much of it appears without any evidence use.

model	coef	ci_low	ci_high	p	perm_null_mean	perm_null_95	perm_p	B_coef	naive_pull_corr	naive_pull_corr_null_mean
pooled	0.43	0.39	0.48	<0.001	-0.00	0.02	<0.001	0.48	0.57	0.44
pooled + input party	0.32	0.27	0.37	<0.001	-0.00	0.02	<0.001	0.41	0.57	0.44
llama_3b	0.12	0.04	0.20	0.004	0.00	0.07	0.004	0.32	0.79	0.78
llama_70b	0.06	-0.06	0.18	0.321	0.00	0.07	0.084	0.48	0.47	0.48
llama_8b	0.65	0.53	0.76	<0.001	0.00	0.09	<0.001	0.35	0.72	0.60
mistral_14b	0.60	0.49	0.72	<0.001	-0.00	0.08	<0.001	0.75	0.34	0.09
mistral_3b	0.66	0.54	0.78	<0.001	0.00	0.10	<0.001	0.33	0.67	0.53
mistral_8b	0.57	0.42	0.72	<0.001	0.00	0.10	<0.001	0.72	0.30	0.15
qwen_32b	0.48	0.40	0.56	<0.001	-0.00	0.06	<0.001	0.56	0.49	0.19
qwen_3b	0.04	0.00	0.08	0.040	0.00	0.03	0.008	0.15	0.50	0.47
qwen_72b	0.43	0.36	0.49	<0.001	-0.00	0.05	<0.001	0.81	0.36	0.02
qwen_7b	0.31	0.18	0.43	<0.001	0.00	0.08	<0.001	0.62	0.30	0.16

Reading: holding the baseline prediction fixed, a one-point shift of the retrieved parties' mean position moves the RAG prediction by 0.43 points (95 % CI 0.39 to 0.48); the permutation null is centred at -0.00. The naive pull correlation is 0.57, of which 0.44 arises with shuffled evidence.

3. Which chunks carry the effect (sample, weighted, descriptive)

Same regression on the sampled texts with the positions of cited vs. uncited and relevant vs. other chunks. The sample is selected on the outcome and lacks the buffer, so read this as descriptive.

comparison	term	coef	ci_low	ci_high	p	n_texts
cited vs. uncited chunks	E_cited	0.35	0.26	0.45	<0.001	638
cited vs. uncited chunks	E_uncited	-0.05	-0.15	0.05	0.355	638
relevant (R_top ≥ 4) vs. other chunks	E_relevant	0.26	0.15	0.37	<0.001	508
relevant (R_top ≥ 4) vs. other chunks	E_irrelevant	0.01	-0.09	0.12	0.777	508

4. Quadrant descriptions (sample)

Quadrants are defined by the errors, so they describe the error cases, not different mechanisms (Q1 / Q2 are selected for large moves, Q4 for none). The last columns test Q1 vs. Q2 with input party and model as controls.

feature	Q1	Q2	Q3	Q4	all (weighted, no buffer)	Q1 vs Q2: log-odds per SD	p (party + model controlled)
mean R_top	2.82	2.99	2.93	2.93	2.98	-0.18	0.085
share chunks R_top ≥ 4	0.35	0.40	0.39	0.38	0.40	-0.15	0.138
mean R_ideo	3.78	3.82	3.78	3.80	3.82	-0.24	0.047
share chunks R_ideo ≥ 4	0.68	0.71	0.69	0.70	0.71	-0.18	0.133
chunks cited (of 5)	2.83	2.61	2.89	2.95	2.84	0.10	0.397
R_top of cited chunks	2.93	3.15	3.06	3.05	3.10	-0.24	0.035
R_ideo of cited chunks	3.81	3.83	3.81	3.86	3.85	-0.16	0.181
share chunks from input's party	0.33	0.26	0.24	0.36	0.30	0.27	0.030
share cited chunks from input's party	0.42	0.31	0.27	0.46	0.36	0.31	0.019
	cited chunks' position − ground truth		0.86	1.13	1.32	0.91	1.10

Share of cited chunks from the input's own party, by input party (Q1 vs. Q2)

input_party	Q1 mean	Q2 mean	Q1 count	Q2 count	position
DIE LINKE	0.72	0.45	24	4	1.85

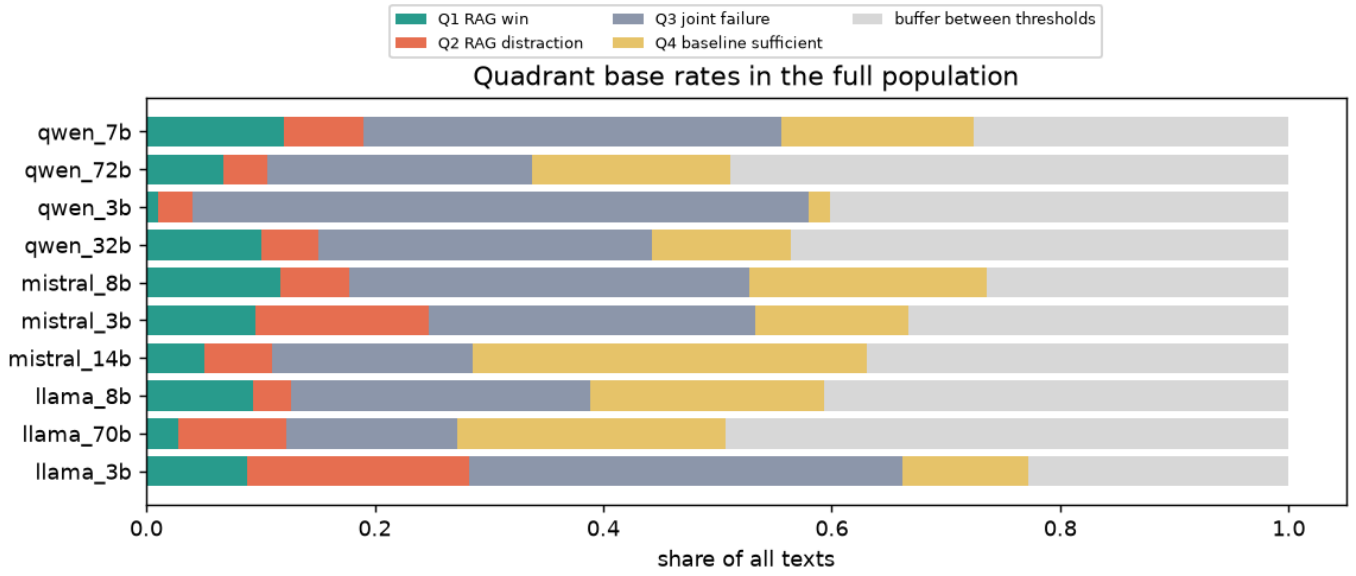
input_party	Q1 mean	Q2 mean	Q1 count	Q2 count	position
BÜNDNIS 90/DIE GRÜNEN	0.24	0.20	33	53	2.83
SPD	0.36	0.21	51	64	3.08
FDP	0.24	0.31	28	24	4.63
CDU	0.28	0.35	20	7	4.95
CSU	0.00	0.00	2	1	5.54
AfD	0.68	0.54	43	48	6.56

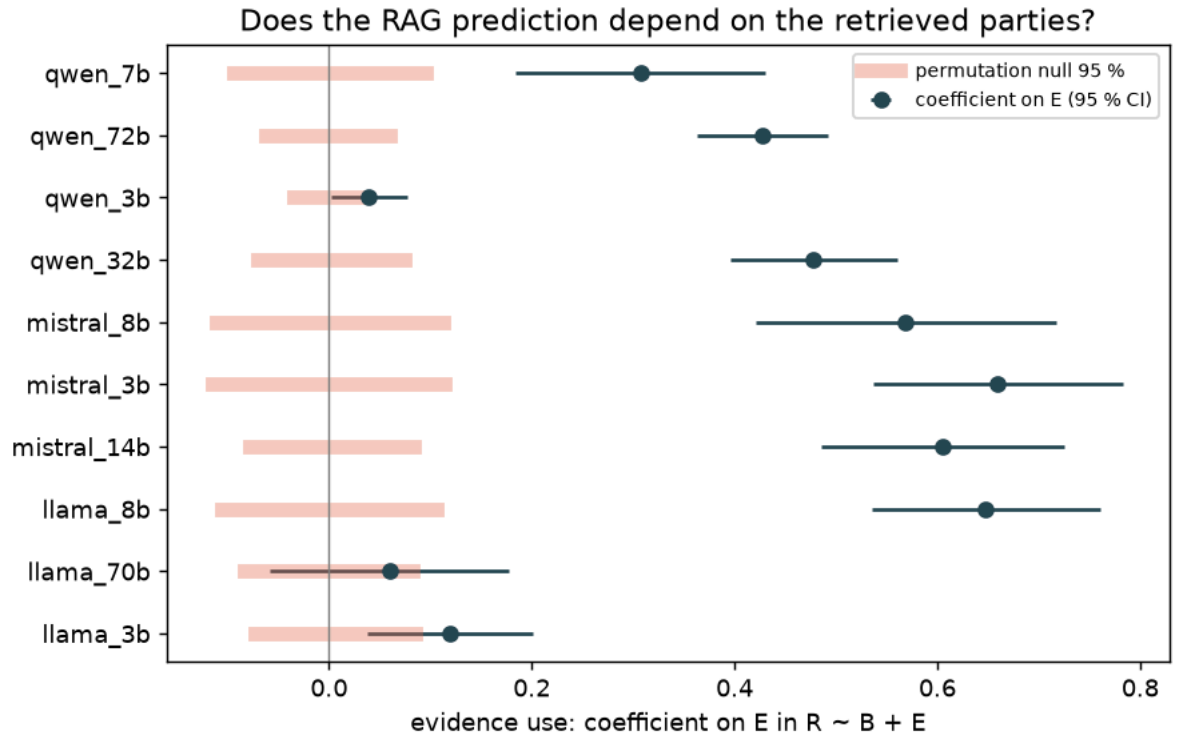
Utilisation: share of chunks cited by label level

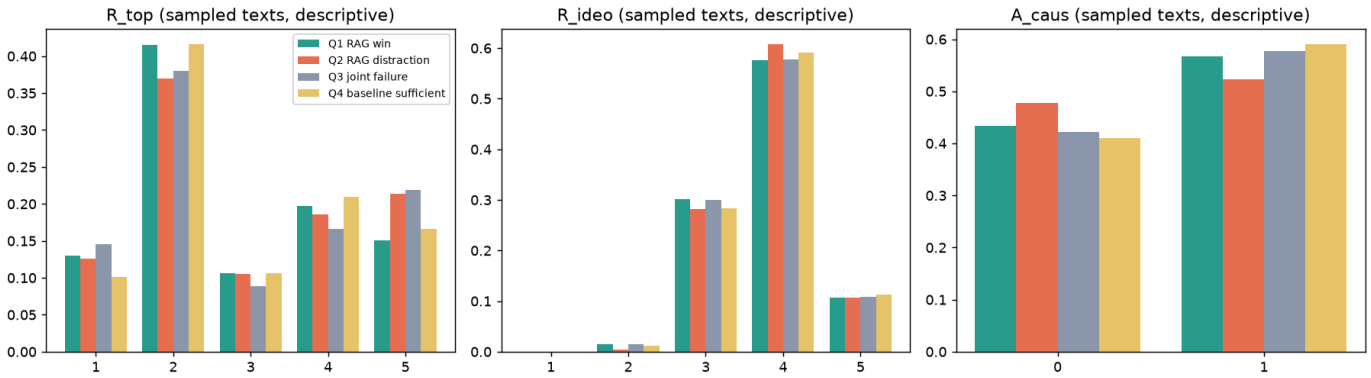
dimension	level	n chunks	cited (weighted)	cited Q1	cited Q2	cited Q3	cited Q4
R_top	1	609	0.45	0.47	0.40	0.48	0.38
R_top	2	1911	0.55	0.53	0.50	0.57	0.57
R_top	3	491	0.62	0.61	0.57	0.59	0.68
R_top	4	917	0.59	0.63	0.54	0.58	0.64
R_top	5	912	0.62	0.63	0.60	0.64	0.65
R_ideo	2	56	0.55	0.59	0.17	0.67	0.27
R_ideo	3	1411	0.55	0.58	0.51	0.58	0.54
R_ideo	4	2848	0.57	0.54	0.54	0.57	0.60
R_ideo	5	525	0.59	0.69	0.49	0.58	0.70

5. Noise floor

Not available: every model has one baseline run (greedy decoding, temperature 0). Re-run the no-RAG baseline once per model and pass it with `--baseline-rerun <model>=<log>`; the share of texts that land in 'Q1' / 'Q2' between two baseline runs is the part of the RAG quadrants explained by run-to-run variation (batching / FP8 non-determinism).


Q1 RAG win
Q2 RAG distraction
Q3 joint failure
Q4 baseline sufficient
buffer between thresholds
Quadrant base rates in the full population
qwen_7b
qwen_72b
qwen_3b
qwen_32b
mistral_8b
mistral_3b
mistral_14b
llama_8b
llama_70b
llama_3b
0.0
0.2
0.4
0.6
0.8
1.0
share of all texts


Does the RAG prediction depend on the retrieved parties?
permutation null 95 %
coefficient on E (95 % CI)
qwen_7b
qwen_72b
qwen_3b
qwen_32b
mistral_8b
mistral_3b
mistral_14b
llama_8b
llama_70b
llama_3b
0.0
0.2
0.4
0.6
0.8
evidence use: coefficient on E in R ~ B + E


R_top (sampled texts, descriptive)
Q1 RAG win
Q2 RAG distraction
Q3 joint failure
Q4 baseline sufficient
R_ideo (sampled texts, descriptive)
A_caus (sampled texts, descriptive)